

BRAC CST+ Project Plan

CMED Health Limited



CMED Health Ltd

Apt # C-5, House # 761, Safura Green,
Satmasjid Road, Dhanmondi, Dhaka-1209

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Project Charter



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Resource Plan

The following resources from CMED will be engaged during the CST+ project to successfully complete the expected deliverables.

Project Management Team

- Moinul Haque Chowdhury
- Shahriar Hossain
- Abdullah Al Mamun

UI/UX Design Team

- Shihab Hasin
- Sayma Yadinin
- Mainul Ishraf

Development Team

- Touhid Mia
- Alamgir Raj
- Chowdhury Arif
- Ariful Haque Anna
- Mahfuz Ibne Alam
- Abdur Razzak
- Shahriar Sohag
- Sanchita Banik
- Tariqul
- Anjarul Raj
- Asifur Rahman
- Ontu Chakrabarti
- Abdur Razzak Riyadh



- Boni Yeasser Soumo
- Toufiqur Rahman Tushar
- Arefinur Rahman
- Mainul Ishraf
- Owares
- Abdullah Al Mamun Shishir

Testing Team

- Habibur Rahman
- Shahriar Hossain
- Ahanab Haque Akib
- Mehedi Hasan
- Pappu

Training Team

- Ashraf Dawood
- Ashiq Khondaker
- Rasel Hossain
- Tamim Hasan
- Reazur Rahman Rahat

Communication Plan

The following personnel will be responsible for the Project Point of Contact from BRAC & CMED.

Project Point of Contact (POC)

A) From BRAC Technology Division:



1. Technical Lead - Muhammad Raquibul Hasan
2. Project Supporting Manager - Arif Hossain
3. Project Coordinator – Tahsin Muhsin
4. Agreement & Contract Focal - Dipayan Datta

B) From BRAC HNPP division:

1. Business Process & Design Focal - Dr. Imran Ahmed Chowdhury & Mohammad Omar Faruque
2. T4D Support - Saiyeed Mashkur Al Hakim & Naimul Islam

C) From CMED:

1. Project Lead – Mr. Moinul Haque Chowdhury
2. Project Coordinator – Mr. Shahriar Hossain & Mr. Habibur Rahman
3. Mobile App Focal – Mr. Alamgir Raj
4. Backend Focal – Mr. Touhid Mia
5. Overall Architecture – Mr. Touhid Mia & Mr. Moinul Haque Chowdhury
6. High-Level Overview – Prof. Khondaker A. Mamun

Escalation Plan

Escalation Point of Contacts:-

1. From BRAC, Technology - Imran Sadik Chowdhury
2. From BRAC, HNPP - Saiyeed Mashkur Al Hakim



3. From CMED – Moinul Haque Chowdhury

Tahsin Muhsin & Shahriar Hossain to communicate & coordinate the project way forward from respective ends.

Work Breakdown Structure (WBS)

Attached in Annex



BRS (Business requirement specification with MVP features)



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Introduction

BRAC was established in 1972, in response to the post-war devastation of Bangladesh's liberation war. Since its founding, BRAC has grown from a small relief operation in a few villages in the country to become the leading non-governmental organization (NGO) in the world. Organizing the poor using communities' own human and material resources, BRAC catalyzes lasting impact, creating an environment in which the poor are empowered to control their own lives. BRAC does this with a holistic development approach geared toward inclusion, using tools like microfinance, education, healthcare, legal services, community empowerment, and among others. BRAC's reach now touches the lives of an estimated 130+ million people, with staff and BRAC-trained entrepreneurs numbering in the hundreds of thousands, a global movement bringing change to Bangladesh and 10 other countries in Asia, Africa, and the Caribbean.

The number of confirmed COVID-19 cases worldwide has crossed 67 million with over 1.54 million deaths in 212 countries and territories so far. A recent study revealed that cases are severely underreported in the country. According to the study findings, approximately 9% of the population of Dhaka city alone (1.62 million people) have been infected at one point in time while the official statistics captured only 66,630 cases. To identify and document the suspected case here requires a digital system by that system project found the live data to take a decision for further management.

Purpose

BRAC is designing an App-based solution that captures Suspected cases (SC), identifies Potential Identified Individual (PII), and follows up PII to ensure proper quarantine procedure is followed. The app will enable us to monitor the community activity at both granular levels as well as in the birds-eye view.

The solution will also ensure the data generated will communicate with other solutions with the possibility of correlating with other medical data like DGHS Solution. Sufficient volumes of COVID-19 data synced with government systems will



lead to hotspot identification, pattern recognition, efficient resource mobilization, and open additional avenues for GO-NGO collaboration. Using a common data platform, (DHIS2) would increase the visibility of interventions, as well as easier access and collaboration.

The Client, The Customer, and other stakeholders

BRAC was established in 1972, in response to the post-war devastation of Bangladesh's liberation war. Since its founding, BRAC has grown from a small relief operation in a few villages in the country to become the leading non-governmental organization (NGO) in the world. Organizing the poor using communities' own human and material resources, BRAC catalyzes lasting impact, creating an environment in which the poor are empowered to control their own lives. BRAC does this with a holistic development approach geared toward inclusion, using tools like microfinance, education, healthcare, legal services, community empowerment, and among others. BRAC's reach now touches the lives of an estimated 130+ million people, with staff and BRAC-trained entrepreneurs numbering in the hundreds of thousands, a global movement bringing change to Bangladesh and 10 other countries in Asia, Africa, and the Caribbean.

Product Users

The users of this product will be as follows -

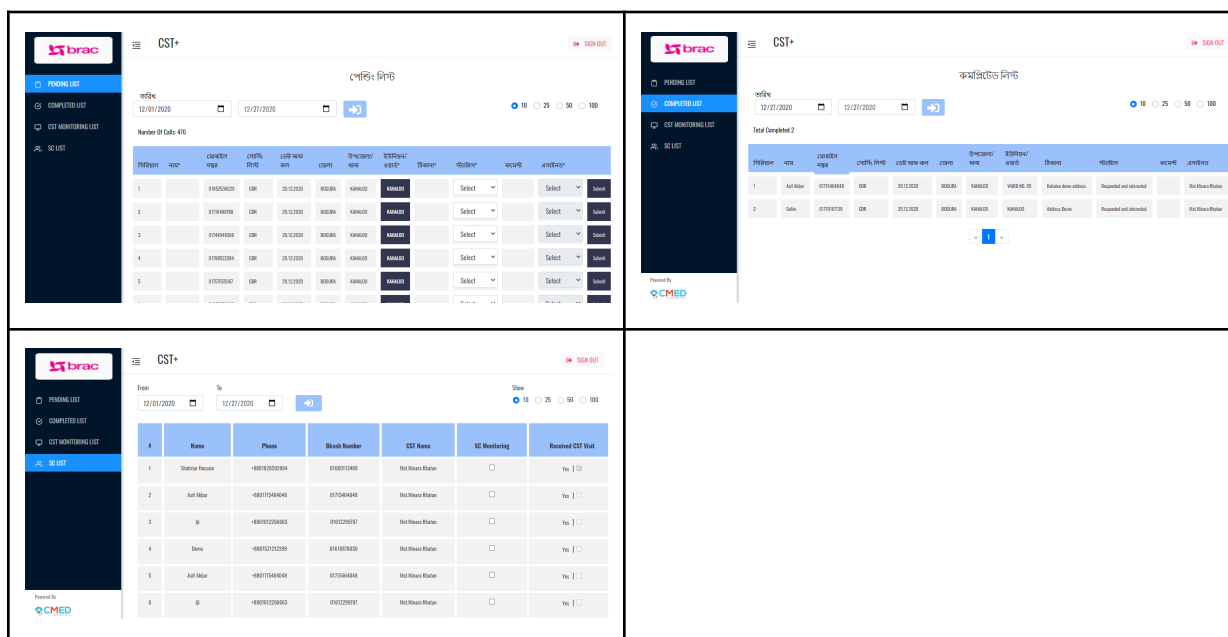
- Community Health Workers
- Area Manager
- District Manager
- Doctor
- Head Office
- UH&FPO of the catchment area



Scope of Work

The development of the following items are within the scope of work -

- Mobile Application for CST
 - ** Screenshot of the UI is added on Annex 2.
- Web Application for Area Manager



The screenshots show the web application interface for CST+ (Community Support Team Plus). The interface includes a sidebar with navigation options: PENDING LIST, COMPLETED LIST, CST MONITORING LIST, and SE LIST. The main content area displays a table of data for CST+.

Screenshot 1 (Top Left): Shows the 'PENDING LIST' for CST+. The table has columns for S/N, Name, Phone, Block Number, CST Name, SE Resulting, and Received CST Visit. The table is filtered by 'Date' (12/21/2020) and 'Show' (10, 25, 50, 100). The table contains 5 rows of data.

Screenshot 2 (Top Right): Shows the 'COMPLETED LIST' for CST+. The table has columns for S/N, Name, Phone, Block Number, CST Name, SE Resulting, and Received CST Visit. The table is filtered by 'Date' (12/21/2020) and 'Show' (10, 25, 50, 100). The table contains 2 rows of data.

Screenshot 3 (Bottom Left): Shows the 'CST MONITORING LIST' for CST+. The table has columns for S/N, Name, Phone, Block Number, CST Name, SE Resulting, and Received CST Visit. The table is filtered by 'Date' (12/21/2020) and 'Show' (10, 25, 50, 100). The table contains 5 rows of data.

Screenshot 4 (Bottom Right): Shows the 'SE LIST' for CST+. The table has columns for S/N, Name, Phone, Block Number, CST Name, SE Resulting, and Received CST Visit. The table is filtered by 'Date' (12/21/2020) and 'Show' (10, 25, 50, 100). The table contains 5 rows of data.

- Web Application for UH&FPO



 UH&FPO



 UH&FPO DASHBOARD

 SC LIST

CST+

UHFPO

 SIGN OUT

 DOWNLOAD

S/L	Telco Source			WORD OF MOUTH APPROACH			CC Approach		TELCO+WORD OF MOUTH APPROACH		Supporting service	Sample Collection	
#	No. of call received	PII screened	SC Identified	Total HHs Visit	PII screened	SC Identified	SC Identified	SC refer to hospital	Total PII	Total SC	Linked with telemedicine consultation	No. SC given consent for Sample Colln	No. of Sample Collected by BRAC
1	16	13	03	21	01	04	03	33	9	5	19	21	26
2	16	13	03	21	01	04	03	33	9	5	19	21	26
3	16	13	03	21	01	04	03	33	9	5	19	21	26
4	16	13	03	21	01	04	03	33	9	5	19	21	26
5	16	13	03	21	01	04	03	33	9	5	19	21	26
6	16	13	03	21	01	04	03	33	9	5	19	21	26
7	16	13	03	21	01	04	03	33	9	5	19	21	26
8	16	13	03	21	01	04	03	33	9	5	19	21	26
9	16	13	03	21	01	04	03	33	9	5	19	21	26

- Web Application for District Manager

District Manager

DM DASHBOARD

CST+

SIGN OUT

From

04/05/2020

To

10/05/2020

Division

Select

District

Select

Upazila

Select

Union

Select

Role

Select

VIEW

Show

100

25

50

100

S/L

S/L	No. of call received	No. of PII assigned to CST	PII screened	SC Identified	Total HHs Visit / CST	PII screened	SC Identified
AM 1	4	1	0	0	40	2	0
AM 1	4	1	0	0	40	2	0
AM 1	4	1	0	0	40	2	0
AM 1	4	1	0	0	40	2	0

- Web Application for Doctor



brac

CST+

From

To

Division

District

Upazila

Union

04/05/2020

10/05/2020

Select

Select

Select

Select

Doctors Dashboard

DOCTORS DASHBOARD

DOCTORS FORM

Name	Phone Number	Symptom Entry	Severity of Symptoms	Collected sample	Consent	Police Lab Submission	Sample result	Source	Advised Medicine	Referral needed	Quarantine	PHF members present	Action
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save
Sc-01	01822333111	Fever, Cough, Sore throat	Severe	N	Y	Radio	Positive	BRAC	Amoxicillin	Y	Y	N	Save

brac

CST+

Search

1

Doctors Dashboard

DOCTORS DASHBOARD

DOCTORS FORM

Name of SC

Phone number / ID

District

Age

Name of CST

Upazila

Gender

Union

brac

CST+

INITIAL CALL WITH CST & SC

EMERGENCY / OTHER CALL

SC CALL IN 14TH DAY

Doctors Dashboard

DOCTORS DASHBOARD

DOCTORS FORM

Symptom entry

If the SC has shortness of breath?

Concomitancies

Is the SC taking anything for these symptoms?

Temperature of SC

Temp °F

Oxygen saturation

SpO₂

Management Plan

Advised Medicine

Save

brac

CST+

INITIAL CALL WITH CST & SC

EMERGENCY / OTHER CALL

SC CALL IN 14TH DAY

Doctors Dashboard

DOCTORS DASHBOARD

DOCTORS FORM

Symptom entry

If the SC has shortness of breath?

Concomitancies

Is the SC taking anything for these symptoms?

Temperature of SC

Temp °F

Oxygen saturation

SpO₂

Management Plan

Advised Medicine

Quarantine maintained by SC

Quarantine maintained by PHF members

What is the current condition of suspected case?

Get any specialized support

Source

- Web Application for Head Office
- Web Application for Super Admin

Out of Scope

1. iOS version of the App.
2. Any feature that is not mentioned in this proposal.



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Features and Functionalities

FEATURES AND FUNCTIONALITIES	ROLE	SCOPE DETAILS
Mobile Application	SK	1. Create Beneficiary 2. Search Beneficiary 3. Create Family Member 4. Create Household Registration Form 5. Create a PII form 6. Create Suspected Case (SC) form 7. View Visit List - View Screened List - View Follow-up List - Create Follow Up Form for 3rd day - Create Follow Up Form for 7th day 8. View Pending Call List (data from Telco Source) 9. View Statistics 10. View Khana visit dashboard 11. View Profile 12. View Service History 13. Offline Sync 14. View Sync Status
Web Dashboard	Area Manager	1. View Telco Data 2. Assign SK 3. View Assigned SK data 4. View SK Household Visit Information / Update SC Monitoring (Check box) 5. Download Report



	Doctor	1. View SC List with follow up history 2. 14th Day Follow up 3. Assign Sample Collector/Lab technician to SC from SC List 4. Download Report 5. CST Link Form (Telemedicine)
	UH&FPO	1. View Union-wise summary information 2. Download Report 3. View key Information 4. Visual Analysis 5. View SC List
	District Manager	1. Download Report 2. View Key Information 3. View Analytics
	Head Office User	1. Download Report 2. View Key Information 3. View Analytics
	Super Admin	1. Download Report 2. View Key Information 3. View SK Household Visit Information 4. Add/Insert User Information & Assign Role 5. Update Master data 6. Update bKash money sent data
Data Integration	System	1. Integration between CST+ and DHIS-2 system through Rest API

MVP Features



SI	Feature & Functionality	Role	Details
1	Mobile App	SK & CC	All features as mentioned above.
2.	Web Dashboard	Doctor	All features as mentioned above.
		Area Manager (AM)	1. Assign SK 2. View Assigned SK data

Non Functional Requirements

Application Feature	Requirements
Security	System security must be maintained as per the best industry practices.
Administrative Settings	- Access Control List (ACL) - User and Role management - Report generation - Audit trail
Source Code Version Controlling	Source code version controlling should be maintained as per the standard best practices (GitHUB/Bitbucket)
Mobile Responsiveness	The system must be mobile responsive for any mobile device (e.g. Mobile, Tablet)
Mobile OS Compatibility	Android version 4.4 or above
Browser compatibility	The system must be compatible with the latest browser versions like- Firefox, Chrome, Edge



Scalability	The system must be scalable enough to accommodate new features, enhancements in the future.
Offline Capacity	Mobile app and Web app should perform with minimum bandwidth
Design compatibility	BRAC brand guidelines should be followed.

Project Deliverables

- CMED will work closely with BRAC HNPP and BRAC Technology, in order to understand and capture the business requirements.
- CMED will take an iterative agile approach to development, releasing usable software in phases that can be tested with the BRAC team.
- CMED will clearly understand the business scope and implement the same in the deliverables.
- CMED will submit a detailed Project Plan including the following documents according to BRAC team requirements:
 - Project charter (Resource plan, Escalation plan, Communication plan, WBS)
 - BRS (Business requirement specification with MVP features)
 - SRS (Software requirement specification with wireframing)
 - SDD (Software Database Design Documentation)
 - Software and System Architecture Document (High Level & Low Level)
 - Quality management plan and test cases
 - Risk management and mitigation plan
 - Source Code- Web and Mobile app (With IP Rights)
 - Deployment architecture document
 - Post-deployment support plan
 - User Manual
 - Training material and Training Plan (For ToT)



- Server requirements (Hardware sizing for Application server, Database server, Master Server, Load Balancer, API server, and Backup storage server)
- API documentation
- System architecture for DHIS-2 Integration with BRAC FCDO CST+ project

Warranty

Minimum 03 (Three) months warranty of all deliverables.



SRS (Software requirement specification with wireframing)



Background

CMED Health Limited is the largest IoT-enabled Artificial Intelligence-driven Cloud-based Digital Healthcare Platform provider in Bangladesh. We have provided digital healthcare services to over 3 million people in the past 4 years in Bangladesh. In partnership with Palli Karma Sahayak Foundation (PKSF), we have already implemented Digital Healthcare services in 51 unions of 48 Upazilas in 36 Districts to establish a referral system and achieve Universal Health Coverage (UHC). We have also developed the National Digital COVID-19 Surveillance System in partnership with the Directorate General of Health Services, which was used by over 18,000 CHCP & Health Assistants to assess COVID risk all across Bangladesh. Following through, we would like to establish an innovative sustainable Union-based General Practitioner (GP) Model utilizing a Digital platform to revitalize the healthcare service paradigm in Bangladesh.

Project Description

The number of confirmed COVID-19 cases worldwide has crossed 41 million with over 1.13 million deaths in 212 countries and territories so far. In Bangladesh, since the detection of the first case on March 8, the virus has spread rapidly across the country, reportedly infecting over 388,569 individuals and claiming 5,660 lives as of October 19, 2020. A recent study revealed that cases are severely underreported in the country. According to the study findings, approximately 9% of the population of Dhaka city alone (1.62 million people) have been infected at one point in time while the official statistics captured only 66,630 cases.

BRAC HNPP's community-based surveillance and active case searching will lead to hotspot identification, ensuring appropriate services and support, and enforcement of healthy behaviors. Through the syndromic surveillance system established in the community using the CHW network of BRAC and the government, BRAC will be able to quickly identify those who meet the criteria of suspected cases. At the Upazila



level, the UHFPO also has access to this information through a live dashboard which helps him/her in informed decision making and resource mobilization.

For the purpose of community-based surveillance, BRAC is designing an App solution-based solution that captures Suspected cases (SC), identifies Potential Identified Individual (PII), and follows up PII to ensure proper quarantine procedure is followed. The app will enable us to monitor the CST program at both granular level as well as bird-eye view. The solution will also ensure the data generated will communicate with other solutions with the possibility of correlating with other medical data. Sufficient volumes of COVID-19 data synced with government systems will lead to hotspot identification, pattern recognition, efficient resource mobilization, and open additional avenues for GO-NGO collaboration. Using a common data platform (DHIS2), would increase the visibility of interventions, as well as easier access and collaboration.

Objective

The objective is to design and develop an Android-based mobile app Application & Multi-layer Web portal for the BRAC CST Project. There will be two parts to the solution. The first part is an easy and user-friendly App (Android) solution for the field operators to collect household and PVF data. The second part consists of a web portal that will be used telemedicine doctors can do follow-up calls and update PFV data, supervisors/government officials to monitor both granular and summary data.

Scope of Work

The development of the following items are within the scope of work -

- Mobile Application for CST
- Web Application for Area Manager
- Web Application for UH&FPO
- Web Application for District Manager
- Web Application for Doctor
- Web Application for Head Office
- Web Application for Super Admin



Hierarchical Structure

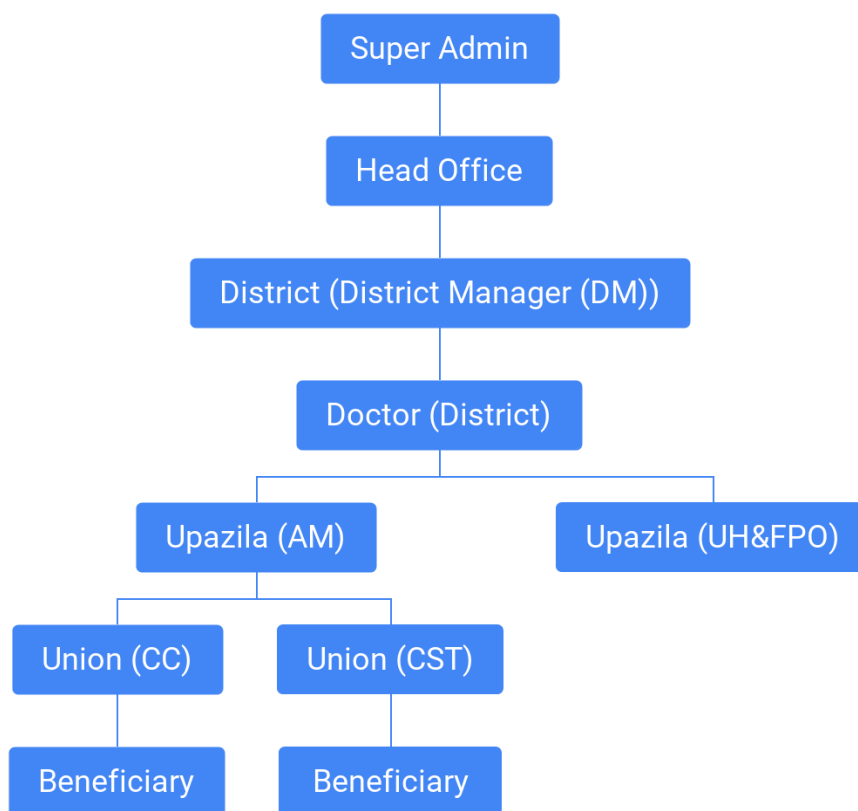


Fig: Project Hierarchy

As per the Project Hierarchy, any node that is above is a parent node to the nodes below them, which are child nodes with respect to that node. Only Super Admin will have the authority to create a user and assign them to a specific role.

Access-control list (ACL)

The access to any specific user role will be controlled via a unique username. All the roles will have different user IDs and passwords. The user ID and password will be provided by Super Admin to the user. To get access to a specific role, a user needs to use their particular user id and password.



SL No	Role	Features	Access panel	Access Level
1a	SK-FIELD	- Create Beneficiary - Search Beneficiary - Create Family Member - Create Household Registration Form - Create a PII form - Create Suspected Case (SC) form - View Visit List - View Screened List - View Follow-up List - Create Follow Up Form for 3rd day - Create Follow Up Form for 7th day - View Pending Call List (data from Telco Source) - View Profile - Offline Sync - View Sync Status	SK Mobile Application	Enabled
1b	SK-CC	- Create Beneficiary - Search Beneficiary - Create Family Member - Create a PII form - Create Suspected Case (SC) form - View Screened List - View Profile - Offline Sync - View Sync Status	SK Mobile Application	Enabled



2	AM	- View Telco Data - Assign CST - View Assigned CST data - SC list of Assigned CST's / Update SC Monitoring (Check box) - Download Report	AM Dashboard	Enabled
3	UH&FPO	- View Union-wise summary information - Download Report - View key Information - Visual Analysis - View SC List	UH&FPO Dashboard	Enabled
4	DM	- Download Report - View Key Information - View Analytics	District Manager Dashboard	Enabled
5	Doctor	- View SC List with follow up history - 14th Day Follow up - Assign Sample Collector/Lab technician to SC from SC List - Download Report - CST Link Form (Telemedicine)	Doctor Dashboard	Enabled
6	HO	- Download Report - View Key Information - View Analytics	Head office user	Enabled
7	Super Admin	- Download Report - View Key Information - View SK Household Visit Information - Add/Insert User Information & Assign Role - Update Master data - Update bKash money sent data	Super Admin	Enabled



Question Form

1. Khana Register Form

Sl no	Questions	Answer	Condition
	Automatic geotagging		
1	উত্তরদাতার নাম	Text	
2	উত্তরদাতার ফোন নাম্বার	Number	
5	পরিবারে কি কোন গর্ভবতী নারী আছে?	Yes/No	
6	পরিবারে কত জন সদস্যের বয়স ৬০ এর উর্ধ্বে?	Number	
7	পরিবারের কতজন সদস্যের উচ্চরক্তচাপ/ ডায়াবেটিস রয়েছে?	Number	
Message	ঘরের বাইরে গেলে মাস্ক পরিধান করুন	Text msg	Yes/No
	ভীড়/জনসমাগোম এড়িয়ে চলুন, একে অন্যের থেকে ৩ ফুট দূরত্ব বজায় রাখুন	Text msg	Yes/No
	কিছুক্ষণ পর পর কমপক্ষে ২০ সেকেন্ড ধরে সাবান পানি দিয়ে হাত ধুতে হবে	Text msg	Yes/No
8	পরিবারের কোন সদস্যের গত ৭ দিনে জ্বর, কাশি, শ্বাসকষ্টজনিত সমস্যা হয়েছে কনি?	Yes/No	
Condition	উত্তর হ্যা হল, পিআইআই (PII) ফরমে চলে যাবে		Condition

2. PII Form

Sl no	Questions (HH)	Answer	Condition
1	পিআইআই-এর নাম	Text	
2	পিআইআই-এর ফোন নাম্বার	Number	
3	পিআইআই-এর বয়স	Number	
4	পিআইআই-এর লিঙ্গ	Male/Female/Others	



5	গত ২৪ ঘন্টায় পিআইআই-এর জ্বর ছিল কি?	Yes/No	
6	পিআইআই-এর শরীরের তাপমাত্রা (ফারেনহাইট)	Number	
Condition	if Q5 is yes but Q6 is < 100 degrees F, go to Q7; otherwise 8		
7	গত ৬ ঘন্টায় পিআইআই কি প্যারাসিটমল/এস/নাপা জাতীয় ঔষধ খেয়েছে কি না?	Yes/No	
8	পিআইআই-এর কি কি উপসর্গ রয়েছে?		
	শুকনা কাশি, মাথা ব্যাথা, ঘ্রাণ না পওয়া, স্বাদ না পওয়া, শরীরে ব্যাথা, শ্বাসকষ্ট, গলা ব্যাথা, ডায়রিয়া, দুর্বলতা, চোখ লাল, স্লেম্মায়ুক্ত কাশি, বমি	Multiple answers	
Condition	শরীরের তাপমাত্রা ১০০ ডিগ্রি ফারেনহাইট বা এর বেশি হলে এবং ৮ নং যে কোন উপসর্গ থাকলে অথবা প্যারাসিটমল/এস/নাপা জাতীয় ঔষধ খেয়েছে, তাপমাত্রা ১০০ নিচে এবং ৮ নং যে কোন উপসর্গ থাকলে এসসি ফরমে চলে যাবে		

3. SC Form

SC Form			
Sl no	Questions	Answer	Condition
1	এসসি-এর নাম	Auto	
2	এসসি-এর ফোন নাম্বার	Auto	
3	এসসি-এর বয়স	Auto	
4	এসসি-এর লিঙ্গ	Auto	
5	এসসি-এর উপসর্গ কবে থেকে দেখা দিয়েছে?	Date	
6	এসসি কি কোন দীর্ঘমেয়াদি রোগে ভুগছেন?		
	হার্টের সমস্যা, উচ্চ রক্তচাপ, ডায়াবেটিস, অ্যাজমা, ক্যান্সার, কিডনী সমস্যা ও অন্যান্য	Multiple answers	
7	এসসি কি গর্ভবতী?	Yes/No	



8	এসসি -এর অক্সিজেনের মাত্রা	Number	
9	আপনি কি করোনা পরীক্ষা করেছেন?	Yes/No	if yes go to q10 / if no go to 11
10	কোথায় থেকে করিয়েছেন	1. BRAC mobile team 2. Govt mobile team 3. Govt Facilities 4. Private health facilities	Radio Button
	করোনা পরীক্ষার ফলাফল কি?	1. Positive 2. Negative 3. Waiting for result	Radio Button
11	আপনি কি করোনা পরীক্ষা করতে ইচ্ছুক?	Yes/No	if yes go to the rest of the questions / no = form close
12	এসসি-কে টেলিমেডিসিন ডাক্তারের সাথে যোগাযোগ করিয়ে দেওয়া হয়েছে?	Yes/No	
13	ডাক্তারের পরামর্শ অনুযায়ী এসসি কে কি হাসপাতালে যওয়ার পরামর্শ দেওয়া হয়েছে?	Yes/No	
14	ডাক্তারের পরামর্শ অনুযায়ী এসসি কে কি ঔষধ প্রদান করা হয়েছে?	Yes/No	
15	পরামর্শ:		
	১।	Yes/No	
	২।	Yes/No	
		Yes/No	



	৩।	Yes/No	
16	CST নাম্বার সম্বলিত স্টিকার প্রদান করা হয়েছে	Yes/No	
17	এসসি-এর বিকাশ নাম্বার		

4. Follow Up Form

Follow up Form (3rd day)			
1	এসসি-এর নাম	Auto	
2	এসসি-এর ফোন নাম্বার	Auto	
3	এসসি-এর বয়স	Auto	
4	এসসি-এর লিঙ্গ	Auto	
Sl no	Questions	Answer	Condition
1	শরীরের তাপমাত্রা (Self Report)	Number	
2	অক্সিজেনের মাত্রা	Number	
3	কি কি উপসর্গ রয়েছে		
	শুকনা কাশি, মাথা ব্যাথা, ঘ্রাণ না পওয়া, স্বাদ না পওয়া, শরীরে ব্যাথা, শ্বাসকষ্ট, গলা ব্যাথা, ডায়রিয়া, দুর্বলতা, চোখ লাল, স্লেথায়ুক্ত কাশি, বমি	Multiple answer	
Condition	If Q9 in SC form is no then ask Q4 again		
4	আপনি কি করোনা পরীক্ষা করেছেন?	Yes/No	if yes go to q5 / if no go to q6
5	করোনা পরীক্ষার ফলাফল কি?	Positive/Negative	
6	আক্রান্ত ব্যক্তি কি সর্বদা বাড়িতে অবস্থান করেছে?	Yes/No	



7	পরিবারের সব সদস্য কি সর্বদা বাড়িতে অবস্থান করেছে?	Yes/No	
8	এসসিকে বিকাশে টাকা দেওয়া হয়েছে	Yes/No	if yes go to q9 / if no go to 10
9	বিকাশে টাকা কবে পেয়েছেন?	Date	
	বিকাশে কত টাকা পেয়েছেন?		
10	পরিবারে অন্য কোন সদস্যের কি করোনার উপসর্গ দেখা দিয়েছে?	Yes/No	
Condition	উত্তর হ্যা হলে, পিআইআই ফরমে চলে যাবে		
Follow up Form (7th day)			
1	এসসি-এর নাম	Auto	
2	এসসি-এর ফোন নাম্বার	Auto	
3	এসসি-এর বয়স	Auto	
4	এসসি-এর লিঙ্গ	Auto	
Sl no	Questions	Answer	Condition
1	শরীরের তাপমাত্রা	Number	
2	কি কি উপসর্গ রয়েছে		
	শুকনা কাশি, মাথা ব্যাথা, ঘ্রাণ না পওয়া, স্বাদ না পাওয়া, শরীরে ব্যাথা, শ্বাসকষ্ট, গলা ব্যাথা, ডায়রিয়া, দুর্বলতা, চোখ লাল, স্লেথায়ুক্ত কাশি, বমি	Multiple answers	
3	আপনি কি করোনা পরীক্ষা করেছেন?	Yes/No	if yes go to q4 / if no go to 5
4	করোনা পরীক্ষার ফলাফল কি?	Positive/Negative	



5	আক্রান্ত ব্যক্তি কি সর্বদা বাড়িতে অবস্থান করেছে?	Yes/No	
6	পরিবারের সব সদস্য কি সর্বদা বাড়িতে অবস্থান করেছে?	Yes/No	
7	এসমিকে বিকাশে টাকা দেওয়া হয়েছে	Yes/No	if yes go to q8 / if no go to 9
8	বিকাশে টাকা কবে পেয়েছেন?	Date	
	বিকাশে কত টাকা পেয়েছেন?		
9	পরিবারে অন্য কোন সদস্যের কি করোনার উপসর্গ দেখা দিয়েছে?	Yes/No	
Condition	উত্তর হ্যা হলে, পিআইআই ফরমে চলে যাবে		



UI Design

Based on the requirements provided by the BRAC team, sample UI design of the mobile application is provided below -

WEB UI Design

Area Manager (AM) Dashboard [Area Manager \(AM\)](#)

Doctor Dashboard and form: [Doctor Dashboard and form](#)

UH&FPO Dashboard: [UH&FPO](#)

District Manager Dashboard: [District Manager](#)

Head Office: [Head Office](#)

App UI Design

CST Role [CST Role](#)

CC Role: [CC Role](#)



App Workflow Diagram

For better view: [CST Plus Flow Diagram.pdf](#)

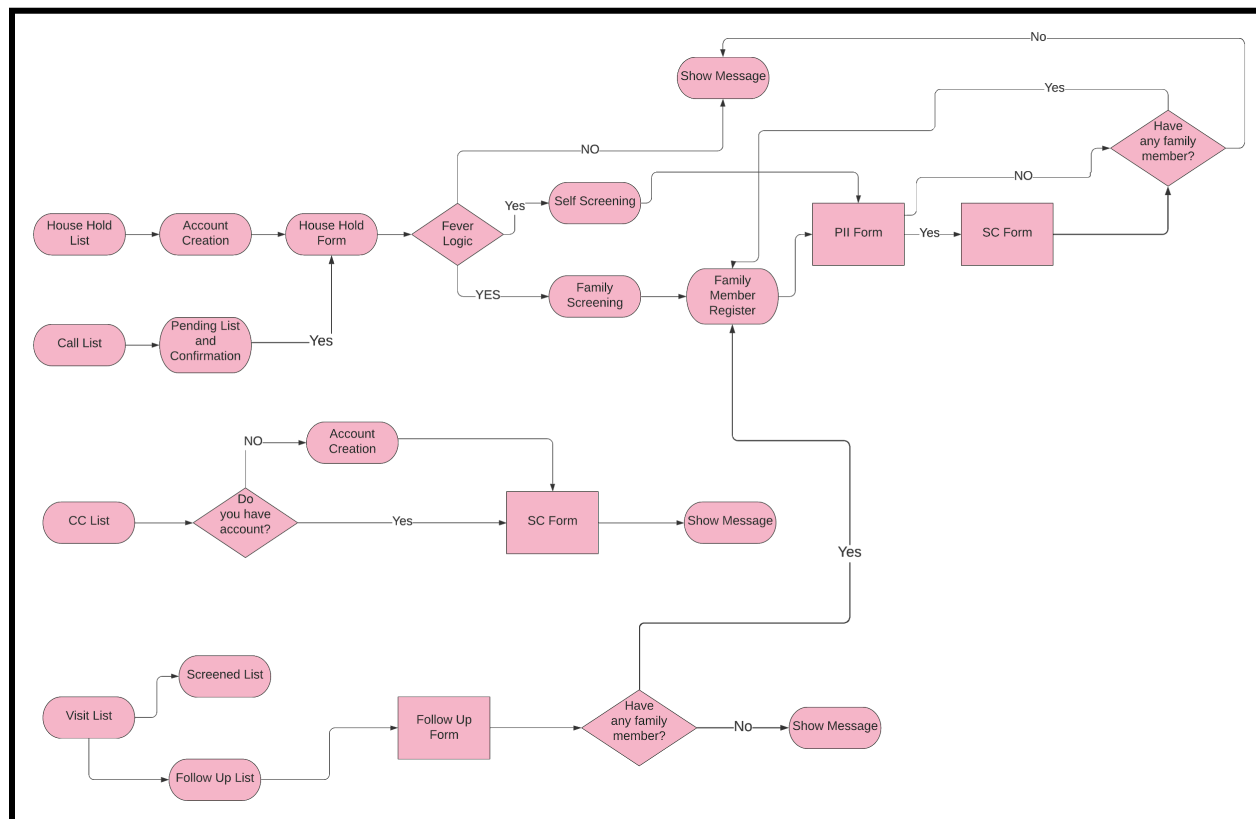


Fig: Flow Chart Diagram

Application Technical Design

All the data will be fed to the National Health MIS DHIS-2. These data can be viewed through a dashboard that will be accessed by implementing partners and the Government of Bangladesh. The web dashboard will be used by different stakeholders to view and monitor the performance of the field-level workers and make decisions based on the collected data. All the data available can be sorted and be downloaded in report format if required. An admin panel will be developed for the

admin user to create/update/delete different roles, send notifications, reminders and have full control over all the available content on the web dashboard.

The technology stack that will be used for the development of the system is given below.

Technology

- Backend (load balanced and auto-scaled using Google App Engine)
- Java Spring Framework
- OAuth2 for authentication
- REST APIs
- Swagger for API documentation
- Redis - Google Memorystore
- Elastic Cloud (Elasticsearch managed service)
- RabbitMQ
- Scheduled ETL jobs
- Materialized views Frontend
- Android: Kotlin, Android SDK

Web front-ends: Angular, Bootstrap4, SASS

UI/UX prototyping

- Adobe XD for Android and Web

Security & Management

- Firewall, DDoS protection, rate limiting, bot management Encryption (Based on the configuration provided by BRAC)
- HTTPS is used for encrypted data transmission using TLS
- Data at rest is stored in encrypted storage at Google Cloud Storage
- Gitlab is used as code repository and for CI/CD pipelines
- Redmine is used for issue tracker
- Scalable application servers

GCP will be used for Load Balancer on Linux, auto scalable instance



- Database server: MariaDB 10.4.16

Cloud SQL MariaDB with daily automatic backups

- Scalable Data storage: Cloud FileStore

Google Cloud Storage

Data integrity, consistency & robustness

Since CMED works with EHR data, ensuring the quality of data is always a high priority to CMED. Data integrity and consistency are mainly maintained by

- Front-end validations
- Business rule validations in server-side
- Utilizing relational database constraints
- Google App Engine automatically terminates faulty servers and re-initiates new servers

Since most of our data is stored in a relational database, all the features offered by relational databases are utilized to maintain data integrity and consistency.

Data backup and recovery

Full database instance snapshots are taken each day automatically and are stored for four weeks in the cloud. Also, a full DB backup is taken each day and downloaded and tested locally.

Privacy

This application manages EHR data, so privacy is a top concern. The following measures were taken to protect the client's privacy:

- User's personal information is not exposed to others (health agent/company admin) without their consent (agreement/acknowledgment/OTP etc.)
- Direct access to the database is not allowed
- Data and backups are stored in the cloud without any human interaction
- Access to servers are regulated by specific roles
- Data backups are downloaded to the local server, and access is restricted to only authorized people
- Headless data (without any personal information) is provided to the developers when extremely necessary to find out the root cause of critical issues



- The CST+ system will be hosted in GCP, which is GDPR ready. GCP aligns the HIPAA risk management program with FedRAMP, SOC 1, ISO/IEC 27001, and NIST 800-53, which are higher security standards that map to the HIPAA Security Rule.

Security

This application is hosted in the GCP cloud and utilizes the security features offered by GCP. Here are some key highlights:

- Encryption is used for data exchange to ensure that information can be safely transmitted over the Internet (HTTPS is used)
- Audit logs and access logs are preserved (who, which data, when, from where)
- Virtual Private Cloud (VPC) was used for the network security features
- Direct access to the application server is not allowed, which is behind a load balancer
- Direct access to the database server is not allowed.
- Each instance is under a GCP security group which allows to specifically open/close inbound and outbound traffic to a specific port
- Passwords are stored in a hash format which has become the industry standard for higher security
- The backend uses Spring security, which is the best in class and offers protection against many security vulnerabilities out of the box.
- Moreover, Cloudflare is used to protect against common attacks like DDoS
- Servers and applications are regularly updated with the latest security patches when available
- All the servers are Linux-based, and all software is used from only the official repositories. So, the risk of viruses and malware is less.

Technical dependencies

The following items are dependent on BRAC.

1. STAGE & PRODUCTION servers delivered in the configuration as requested
2. The domain will be provided by Brac with the delivery of Stage Server
3. VPN Access or Direct server access will be provided by Brac (Not Decided yet)



List of deliverables

Project Plan including the following documents:

1. Project charter (Resource plan, Escalation plan, Communication plan, WBS)
2. BRS (Business requirement specification with MVP features)
3. SRS (Software requirement specification with wireframing)
4. SDD (Software Database Design Documentation)
5. Software and System Architecture Document (High Level & Low Level)
6. Quality management plan and test cases
7. Risk management and mitigation plan
8. Source Code- Web and Mobile app (With IP Rights)
9. Deployment architecture document
10. Post-deployment support plan
11. User Manual
12. Training material and Training Plan (For ToT)
13. Server requirements (Hardware sizing for Application server, Database server, Master Server, Load Balancer, API server, and Backup storage server)
14. API documentation
15. System architecture for DHIS-2 Integration with BRAC FCDO CST+ project



Server requirements

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Server Configuration

Server Types	STAGE, PRODUCTION
Number of servers for each type	Total 6 (STAGE 3 & PRODUCTION 3)
Operating System	Linux
OS Name:	CentOS
OS Version	8 (64 bit)

Server: Production				
Sl No	Servers (VMs)	CPU Core	RAM	SSD
1	Application Server #1 (Linux)	8	16GB	100GB
2	Application Server #2 (Linux)	8	16GB	100GB
3	Load balancer for Application Servers (Linux)	GCP Cloud Load Balancing		
4	Database Server - Master Node (Linux) Cloud SQL MariaDB	16	32GB	500GB
5	Cache & Messaging server - Redis	GCP Memorystore Redis		
6	Cache & Messaging server - ElasticSearch	GCP Elastic Cloud (Elasticsearch managed service)		
7	Cache & Messaging server- RabbitMQ	GCP RabbitMQ		
8	1. Application Backup - 1 Backup Snapshot Daily (7 Days Retention Period) 2. Database Backup - Total 4 Backup Snapshot Daily with 3 Peak Hour & 1 Off-Peak Hour (7 Days Retention Period)	GCP Snapshot		



Server: Stage				
Sl No	Servers (VMs)	CPU Core	RAM	SSD
1	Application Server #1 (Linux)	4	8GB	50GB
2	Application Server #2 (Linux)	4	8GB	50GB
3	Load balancer for Application Servers (Linux)	GCP Cloud Load Balancing		
4	Database Server - Master Node (Linux) Cloud SQL MariaDB	8	16GB	250GB
5	Cache & Messaging server - Redis	GCP Memorystore Redis		
6	Cache & Messaging server - ElasticSearch	GCP Elastic Cloud (Elasticsearch managed service)		
7	Cache & Messaging server- RabbitMQ	GCP RabbitMQ		
8	1. Application Backup - 1 Backup Snapshot Daily (7 Days Retention Period) 2. Database Backup - Total 4 Backup Snapshot Daily with 3 Peak Hour & 1 Off-Peak Hour (7 Days Retention Period)	GCP Snapshot		



Software and System Architecture Document (High Level & Low Level)



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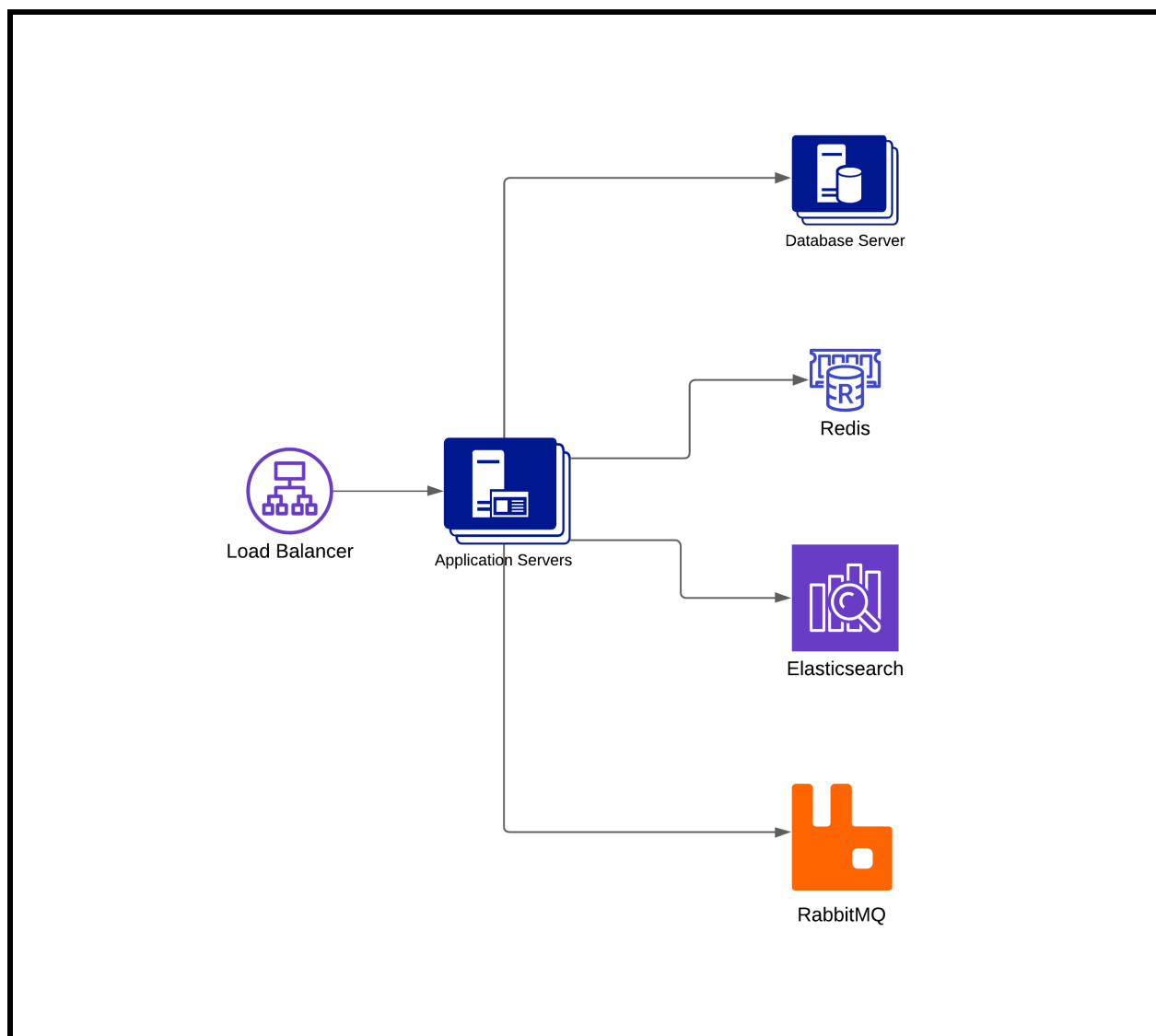
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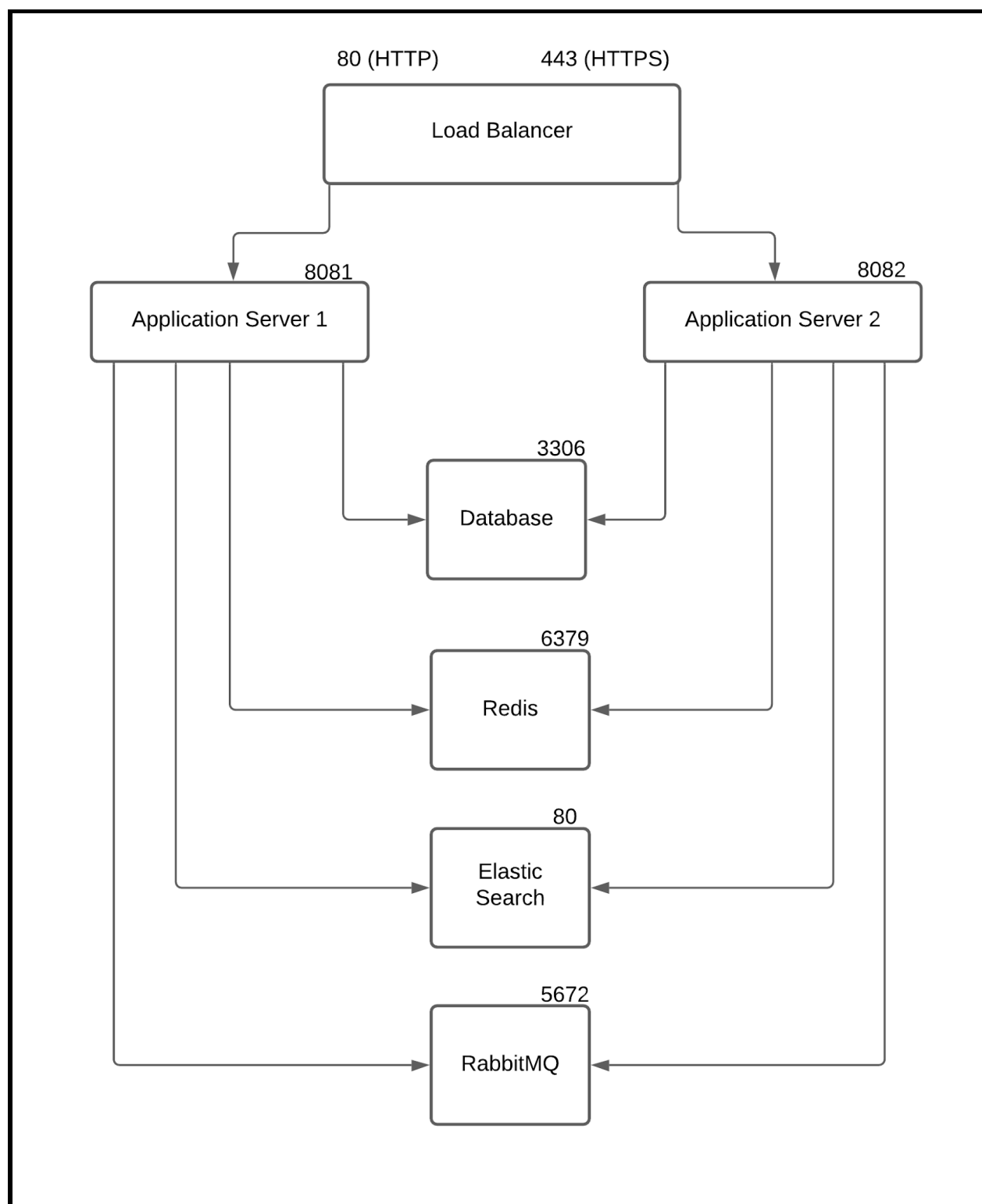
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High Level Design (HLD)



Low Level Design (LLD)

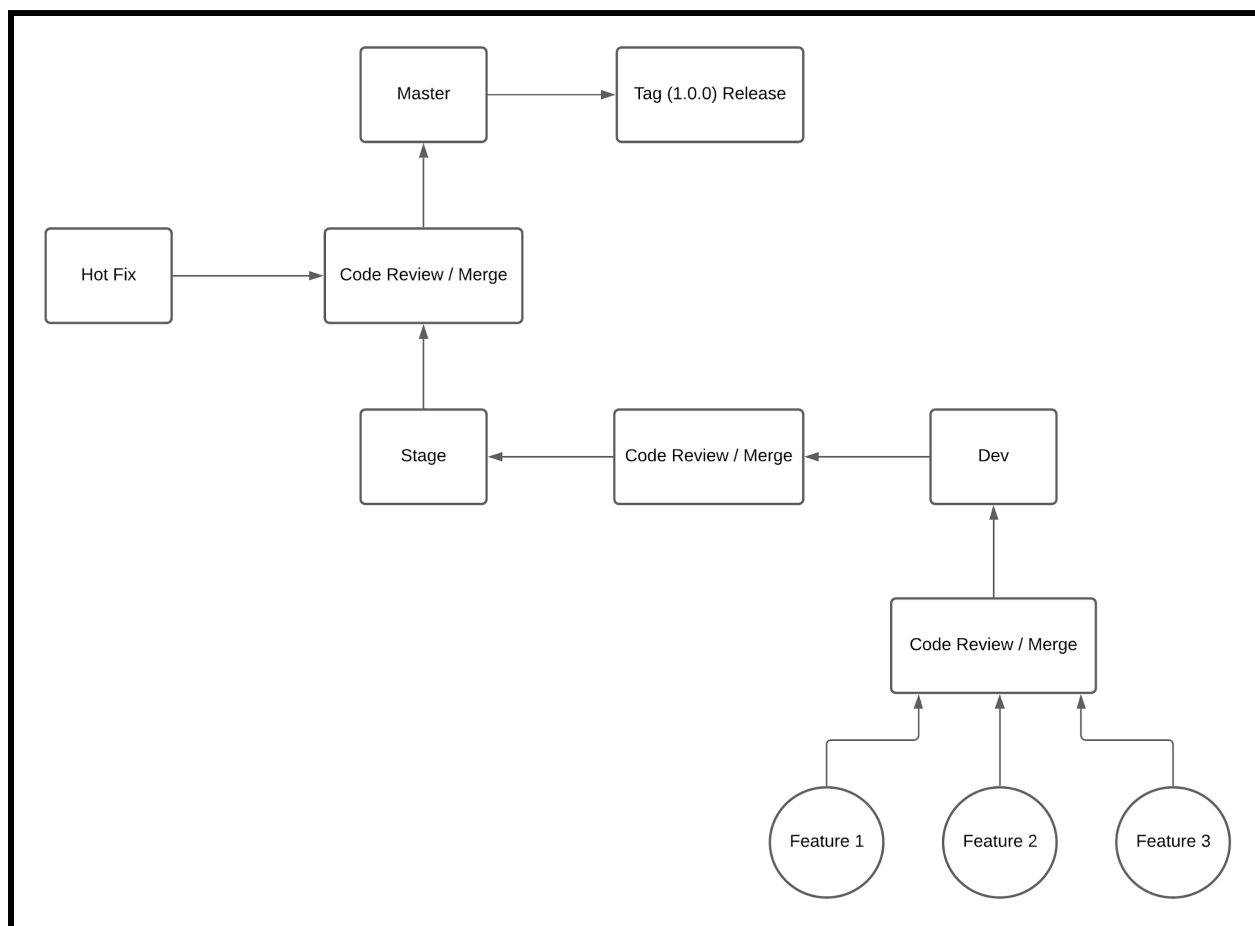


Communication Matrix

Source	Destination	Port Number
Load Balancer	Application Server 1	8081
Load Balancer	Application Server 2	8082
Application Server 1	Primary Database	3306
Application Server 2	Primary Database	3306
Application Server 1	Redis	6379
Application Server 1	Elastic Search	80
Application Server 1	RabbitMQ	5672
Application Server 2	Redis	6379
Application Server 2	Elastic Search	80
Application Server 2	RabbitMQ	5672



Tagging, branching & Source code version controlling



Deployment Architecture Document



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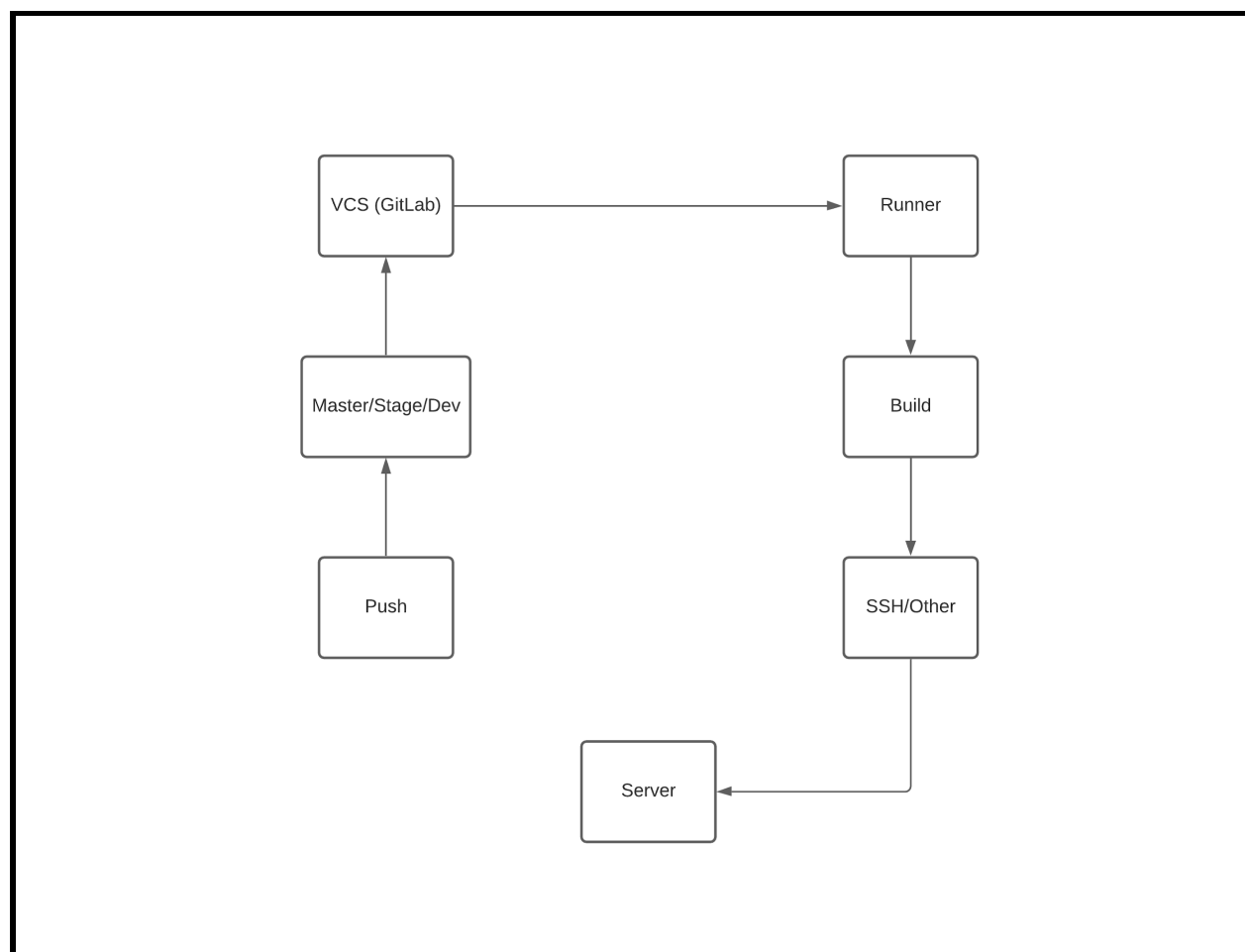
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Continuous Integration (CI) & Deployment Process



Notes:

- For details of how code traverse from repository to dev, stage & master (production), please refer to the section - Tagging, branching & Source code version controlling
- For recovery or rollback, our first approach is to give a hotfix directly to the Master (Production) but if this doesn't solve the issue then we can revert the commit & rollback to the previous version as we are already taking a tag for version release.



General Constraints

- The Internet connection is a constraint for the web application. Since the application fetches data from the database over the Internet, it is crucial that there is an Internet connection for the application to function.
- Android version over or equivalent to 5 will be acceptable for the application.
- 2GB Ram for the tab or smartphone is preferable for running the application.
- The system needs minimum 3G speeds to be workable & sync data.

Assumptions to be Considered

- BRAC must ensure detailed brief requirements before starting the development of the project.
- Stage and Production server will be provided by the BRAC with VPN Access, DEV server will be provided by the CMED.
- The domain will be provided by the BRAC.

System Performance Process

The system should be able to handle a minimum of 1000 users from mobile & web applications simultaneously (According to the Production Server requirement we provided). The system should also be able to deal with a good number of users at the same time. Additionally, the database of the system should be able to handle multiple users at any period.

App Security Implementation Process

This section is referred to the Security part of the proposal.

Exception Handling Process

Primary Feedback:

- We are writing different levels of logs (e.g. error and warn) of the application so that we can check the log later and reproduce the errors/warnings to fix them very quickly.
- If the application throws an exception for any particular feature, the application will continue for other features if the features are independent. In case the server faces any major error, it will be shut down gracefully.

Secondary Feedback:

- We followed standard coding practice and tried to avoid code smell which is more maintainable and readable code.
- We applied layered/modular architecture.





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Quality Management Plan and Test Cases



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Test Plan Overview

Assumptions

1. Walkthroughs of Project Requirements and Detail Design documents are done.
2. Test Environment – the QA environment will be the designated area for in-depth functional testing. It will be ‘frozen’ and will be modified or updated only in coordination with the Test Teams.
3. Database access: The tester will have read/write access to QA/Stage database instances.
4. The tester has all necessary software applications installed on his/her PC and has read/write access to shared network drives as required.

Scope of Testing

In Scope

All the modules, features, and functionalities of the application will be under the testing scope.

Role: CC, CST

- Create Beneficiary
- Search Beneficiary
- Create Family Member
- Create Household Registration Form
- Create PII form
- Create Suspected Case (SC) form
- View Visit List
- View Screened List
- View Follow-up List
- Create Follow Up Form for 3rd day
- Create Follow Up Form for 7th day
- View Pending Call List (data from Telco Source)
- View Profile
- Offline Sync
- View Sync Status

Role: Area Manager

- View Telco Data



- Assign CST
- View Assigned CST data
- SC list of Assigned CST's / Update SC Monitoring (Check box)
- Download Report

Role: Doctor

- View SC List with follow up history
- 14th Day Follow up
- Assign Sample Collector/Lab technician to SC from SC List
- Download Report
- CST Link Form (Telemedicine)

Out of Scope

- View Statistics
- View Khana visit dashboard
- View Service History

Testing Prerequisites

Documentation Requirements

Below is the list of general documents to commence the testing process.

- Project management plan
- Requirement specification
- Business rules and features list
- Database design document
- Sample data and report
- Traceability matrix
- User Acceptance Test (UAT) requirements and test cases.



Testing Environment Requirements

The matrix below contains itemized environmental requirements.

Platform	System & Application Software
Database	MariaDB 10.4.16
Operating System	Windows, Linux, Mac, Android
Browser	IE, Mozilla Firefox, Safari, Google Chrome, Opera or any html5 supported browser
Test Devices	Tab/Mobile Android 5.0+ Devices (Only Portrait Orientations will be supported)
Test Type	Manual Testing will be executed on both Web & Android Applications for functional testing Automation Testing will be executed for API Load Testing
Reporting Tool	Google Sheet, Redmine
Automation Tool	Swagger for API Testing Apache Jmeter for API Load Testing
Bug Reporting Tool	Redmine
Image Capture	Snipping tool

Entrance Criteria

The following criteria must be met before testing can begin:

Entrance Criteria Description	Status
The project Test Plan has been approved.	
System Test cases have been approved	
Environmental Requirements have been satisfied.	
Documentation Requirements have been satisfied.	
Staffing and Training Requirements have been met.	
Technology and Tool Requirements have been met.	



Prerequisite Walkthroughs and Reviews have been completed.	
Test Environment is set up with required data and tools	

Test Approach

All the Test design approaches will be started after the baseline of SRS (Software Requirement Specifications). Test engineers will write test cases deriving from software requirement specification (SRS). If the development process follows Agile Method, then test cases will be executed accordingly, depending on developed modules. Each deployment has to go through Smoke testing to test the minimum required functionalities of over-developed modules manually. Rigorous functional tests will commence once the smoke test is passed. All the defects found will have to go through under defect life cycle including retesting and regression testing. Non-functional tests will be planned and executed according to the project life cycle and requirement need once the functional test is completed.

User Acceptance Test or UAT can be performed only when all the high and medium priority defects are closed in the QA environment and all the test cases within the scope are passed.

Test Types

Below are the types of tests the testing team will execute.

- Smoke Testing** – After any deployment test engineers will run the smoke test to evaluate whether or not the database or process is generally ready for testing. Smoke tests will be used in the pre-QA test cycle(s). Functional tests will be started when the system will pass the smoke test cycle. Some smoke test cases may be repeated in the QA test cycle(s).
- Functional Testing** – The purpose of functional testing is to validate the behavior of the software from an end-user perspective. Inputs and outputs are compared to the expected output. Here the testers will validate all the functionalities of the software system whether it works as required and expected. Functional tests will be the primary basis for the QA test cycle(s).
- Retesting and Regression Testing**– Execution of test cases specifically designed to retest failure conditions that have previously been identified within the application. Retesting and regression testing can take place in any test cycle (QA or UAT) when faults were encountered during an earlier cycle within that group. All scope of regression tests will be identified according to the traceability matrix and scope of the features. Each time whenever any change occurs in an application like defect fixing, new features adding retesting and regression tests will be taken place.



- **Unit Testing** – Unit testing will be done by the development team based on the project requirements and scope. Unit test results will be used as entry criteria for commencing functional testing.
- **Data Validation testing** – All the application data will be validated with the database by the CRUD testing approach. Testers with a good understanding of SQL language and software requirements will create their own queries to test the database and validate it with the application.
- **API integration testing** - API testing will focus on whether API is doing the task it is supposed to do. Test cases for the API validation will contain different input parameters and verify the response using Swagger. The data, response code, message, etc should be correct for each set of input parameters. The API should return correct HTTP error codes like 200 for valid and 400 for invalid input parameters.
- **Browser Compatibility Testing** - The purpose of cross-browser testing is to provide a consistent behavior and experience across all browsers, devices, and platforms. Deciding the number of browsers to be considered for cross-browser testing will be depending upon project requirements such as popular web-browsers such Mozilla Firefox, Opera, Chrome, Internet Explorer 5 and above, Safari, etc. All the app functionalities should be tested in the Mobile version of the application.
- **Mobile Android Version Compatibility Testing** - Testers will check the compatibility of the app with different os versions of Android mobile & tab to ensure the responsiveness of the app.
- **Performance and load Testing** – Using the automation tool Apache Jmeter, testers will evaluate how fast a system responds to a specific scenario and how many responses it can handle simultaneously.
- **Recovery Testing**- Under Recovery testing testers will check whether the application or software behaves and recovers correctly when there is a failure.

Risks and Contingencies

Risk	Priority	Impact	Mitigation Plan
Provide all the necessary hardware and software.	High	Delay in testing.	

Test Deliverables

Work products and activities generated out of the testing processes are as follows. All the test Revision History documents will be updated according to change in the project plan and



requirements.

- **Test Plan** (this document) – Provides information on the testing approach, methods of validation, and the measures & dimensions to be tested. Test Cases give detailed coverage of the testing.
- **Test Cases** – Identify the specific situations/circumstances, positive and negative, to be tested and ensure detailed coverage for the goals set out by the Test Plan
- **Test Results** – Document the results of each Test Case. In the case of failures (full or partial), document the details, symptoms, etc surrounding the failure so as to facilitate resolution by the DBA and developers.
- **Defects** – Defects and potential issues (questions) found during execution of the Test Cases will be logged in Redmine, as they are discovered.

Test Cases, Procedures & Results

The test cases template for testing is attached –

Project Name:											
Requirement DOC #											
Environment (Hardware & Software Configuration)											
Test Case ID	Type	Module	Sub Module	Pre-Condition	Test Scenario	Description	Test Data	Expected Result	Actual Result	Pass/Fail Status	Defect ID

Status Reporting

Defect Review meetings (daily): these will be scheduled by the Test Manager and include key members from the Development and Test teams and Project Management to discuss new issues found, update existing issues, and review any important issues bearing upon fault-resolution efforts.

Test Status report (daily): this will be written and communicated by the Test Manager to the Project Manager and stakeholders. It will contain testing highlights, activities for the next week, and identify important outstanding issues. It will also include key testing-project metrics such as % of testing completed, number of defects found for each QA cycle, number



of defects based on priority level, etc.

Defect Tracking Procedures

Redmine will be used to track defects. The following procedure will be followed when tracking issues documented in the defect tracking database.

- The Tester who identifies the issue will log all pertinent information into the Defect Tracker fault tracking system with status **NEW**.
- The fault will be assigned to the individual responsible for addressing the fault.
- New issues will be reviewed during the defect review meetings so as to ensure effective communication of the issue and obtain 'buy-in' from all parties involved.
- The individual assigned to attend to the defect will change the defect's status to **IN PROGRESS** when he/she begins working on the fix.
- The individual assigned the defect will then change the Status to **Ready for Testing** when the fix is completed and the code fix has been moved to the testing environment. They will also add pertinent fix information to the defect record.
- In the subsequent QA cycle, the Testers will review/retest all of his/her defects marked as **Ready for Testing**, to ensure the issue has been corrected.
- Once a defect has been tested and deemed to be fixed, the tester will set the status of the defect to **QA Verified**. Lead Tester will change the issue status to '**CLOSED**' once its completely resolved.
- If the fix is inaccurate, the tester will set the status of the defect to, '**REOPENED**'.
- An issue that is in '**REOPENED**' status will then be reviewed by the assigned individual and will be corrected and marked as '**Ready for Testing**', adding additional relevant information to the fault.
- This flow will continue as necessary until issues have been resolved & status changed to '**CLOSE**'.
- Testers will be responsible for following through with the issue until corrected.
- Testers will be responsible for updating the defect tracker (Redmine) with timely information about the progress of a defect. If the defect is resolved then the tester set the status of the defect to '**CLOSED**'.

Schedule

Testing Activity	Scheduled Start Date	Schedule End Date	Assigned To
Test Plan	Refer to Project Plan Schedule	Refer to Project Plan Schedule	Habibur Rahman
Test Cases	Refer to Project Plan Schedule	Refer to Project Plan	Mehedi Hasan Shahriar Hossain,



		Schedule	
Functional Testing	Refer to Project Plan Schedule	Refer to Project Plan Schedule	Mehedi Hasan Shahriar Hossain,

Test Resources and Roles

Resource Type	Resource Title	Resource Name	Date Required	Responsibilities
Lead Tester	Lead QA Engineer	Habibur Rahman	Refer to Project Plan	• Prepare test plan • Ensure related support during test execution • Monitor execution process • Communicate with other stakeholders and review the bugs.
Test Engineers	QA Engineer	Shahriar Hossain, Mehedi Hasan,	Refer to Project Plan	• Prepare test case for the project • Execute security testing, browser compatibility testing and another testing • Execute functional and regression testing • Prepare test case for the project • Execute functional and regression testing • Develop test environment
UAT Team (Clients)	End-User	Client representatives	Refer to Project Plan	• Perform User Acceptance Test



UAT Plan

We shall accommodate extensive UAT testing on each feature of each module incorporating representatives from the client.

User Acceptance Criteria

Will be provided by the end-user.

Exit Criteria

All the testing activities listed below must be successfully completed before moving on to the next level of testing.

Exit Criteria Description	Status
All Test Cases have been executed & results validated.	
All 'High' and 'Medium' severity level defects found during System testing have been fixed and successfully retested.	
Any unresolved 'Low' severity level defects found during System testing have been documented and are being addressed by Systems Development.	
Acceptance testing criteria has been exercised and approved by User Group	



Risk Management and Mitigation Plan



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RISK	IMPACT	FREQUENCY	MITIGATION
Communication Barriers	H	M	Due to the location difference of the program and the developer could be challenging. Establish a proper communication channel amongst stakeholders and ensure all parties utilize the mode of communication for their inputs.
Scope inflation during project lifecycle (scope creep)	H	M	The constant involvement of stakeholders and developers.
Resource turnover (Key personnel leaving the project; taking critical information)	H	L	- Increased collaboration and information sharing on the team. - Creating backup pool (resources).
Incomplete or conflicting requirements	H	M	Dedicated Project Manager to clarify from business owners in a quick escalated method.

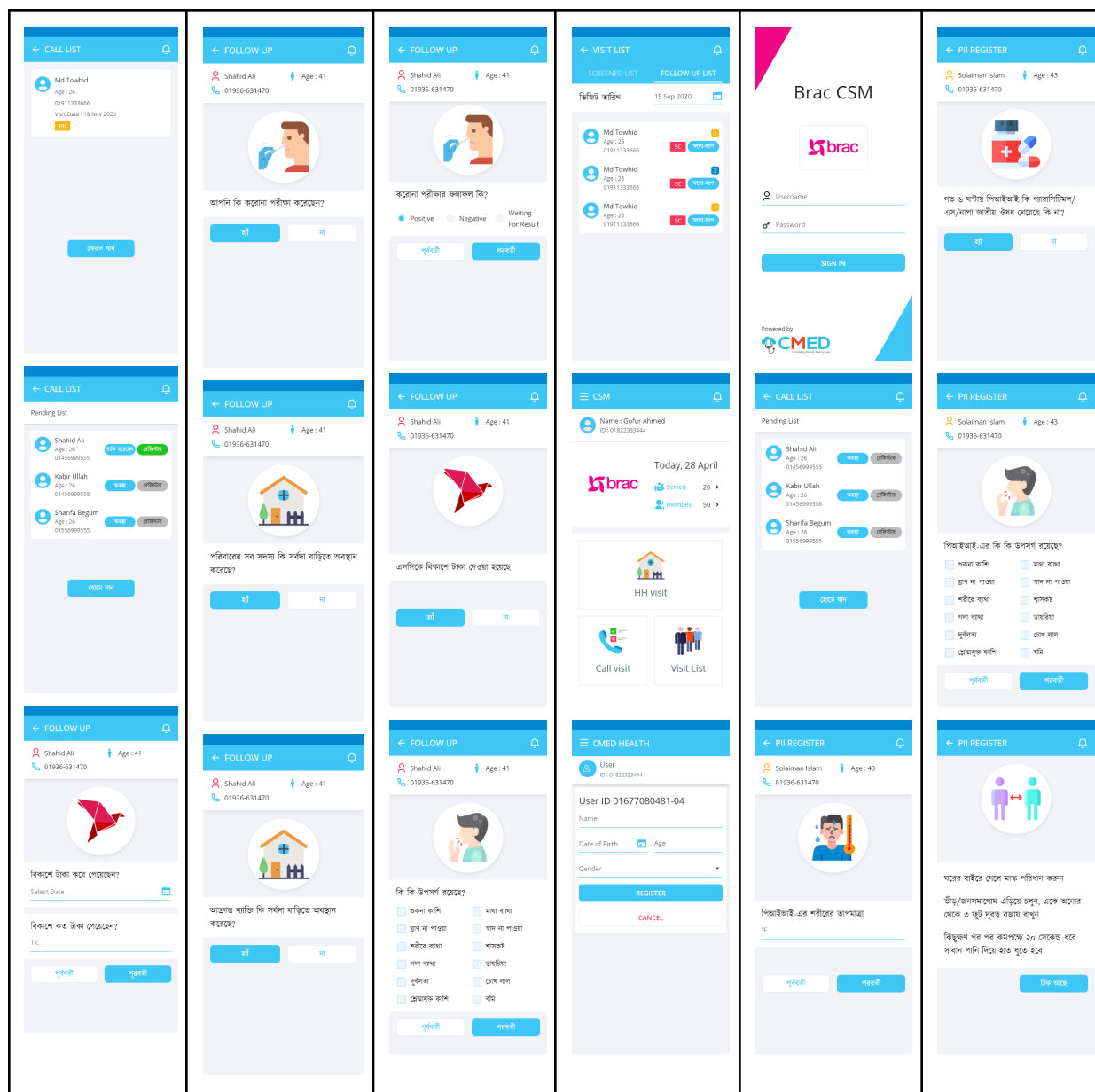


Productivity burnout (long project timelines, sense of urgency)	M	H	Short iterations, right people on team, coaching, and team development.
Overburdened managers	H	H	Ensure there is a degree of flexibility in project plans that anticipates high volumes of work that could pose as impediments to progress related to the project.
Effort and Scheduling Variance	M	H	Get the team more involved in planning and estimating. Get early feedback and address potential slips directly with stakeholders.
Data Integrity Risks	H	M	Ensure proper validation processes in place, clean up and back up data regularly, and audit changes to data and permissions.
Data Security Risks	H	M	Multi-factor authentication and restriction of administrative privileges.



Annex

1. Work Breakdown Structure (WBS): [WBS Pdf Link](#)
2. APP UI Screenshot Field Role



FOLLOW UP Shahid Ali Age: 41 01936-631470 শরীরের তাপমাত্রা পূর্ববর্তী পরবর্তী	FOLLOW UP Shahid Ali Age: 41 01936-631470 এগুপি -এর অস্ত্রিলেগের মাত্রা SPO₂ পূর্ববর্তী পরবর্তী	FOLLOW UP Shahid Ali Age: 43 01936-631470 পরিবারের অন্য কোন সদস্যের কি কয়েমার উপসর্গ দেখা দিয়েছে? হ্যাঁ না	CSM User ID 01677080481-04 Name Date of Birth Age Gender REGISTER CANCEL	PII REGISTER Solaiman Islam Age: 43 01936-631470 গত ২৪ ঘণ্টায় শিঙাইআই-এর ঘুর ছিল কি? হ্যাঁ না	PII REGISTER Solaiman Islam Age: 43 01936-631470 পরিবারের অন্য কোন সদস্যের গত ৭ দিনে ঘুর, কানি, শ্বাসকষ্টজনিত সমস্যা হয়েছে কিনা? হ্যাঁ না
CMED HEALTH Serve Now CMED ID/Phone Number 0167-70804 Kobir Ullah Age: 41 Shahid Ali Age: 41 Sharifa Begum Age: 34 Md Towhid Age: 26 Create a new user under this Mobile Number	Brac CSM Powered by CMED FOLLOWUP LIST Md Towhid Age: 26 01911333666 Visit Date: 18 Nov 2020 Follow up: 3rd day CREATE NEW	VISIT LIST SCREENED LIST FOLLOW-UP LIST ভিজিট তারিখ 15 Sep 2020 Md Towhid Age: 26 01911333666 Md Towhid Age: 26 01911333666 Md Towhid Age: 26 01911333666 CREATE NEW	SCREENED LIST Md Towhid Age: 26 01911333666 Visit Date: 18 Nov 2020 CREATE NEW	CSM Serve Now CMED ID/Phone Number 0167-70804 Shahid Ali Age: 25 Kobir Ullah Age: 41 Sharifa Begum Age: 39 Md Towhid Age: 26 Create a new user under this Mobile Number	CMED HEALTH Serve Now CMED ID/Phone Number 0167-70804 Kobir Ullah Age: 41 Shahid Ali Age: 41 Sharifa Begum Age: 34 Md Towhid Age: 26 Create a new user under this Mobile Number



CC Role

← HH REGISTER Solaiman Islam Age: 43 01936-631470  পরিবারে কত জন সদস্যের বয়স ৬০ এর উপরে? No. 60+ members পূর্ববর্তী পরবর্তী	← HH REGISTER Solaiman Islam Age: 43 01936-631470  পরিবারের কোন সদস্যের গরু ৭ দিনে জ্বর, কাশি, শ্বাসকষ্টজনিত সমস্যা হয়েছে কিনা? হ্যাঁ না	← HH REGISTER Solaiman Islam Age: 43 01936-631470  পরিবারে কি কোন গর্ভবতী নারী আছে? হ্যাঁ না	← HH REGISTER Solaiman Islam Age: 43 01936-631470  মুদ্রাধারক পিতামহ যে মুদ্রাধারক ৫মুদ্রা নিয়ে মুদ্রাধারক জন্ম পরিবারিক মুদ্রাধারক ৫মুদ্রা নিয়ে বহুবারের পরিবারের সদস্য মুদ্রাধারক	← PH REGISTER Solaiman Islam Age: 43 01936-631470  গত ২৪ ঘণ্টায় পিআইআই-এর জ্বর ছিল কি? হ্যাঁ না
← HH REGISTER Solaiman Islam Age: 43 01936-631470  ঘরের বাইরে কোন মাক পরিদর্শন করুন কীট/অণুজীবমুক্ত এড়িয়ে চলুন, একে ঘরের বোকে ও খুঁচু খুঁচু বজায় রাখুন কিছুক্ষণ পর পর কমপক্ষে ২০ সেকেন্ড ঘরে সাবান পানি দিয়ে হাত ধুতে হবে পূর্ববর্তী পরবর্তী	← HH REGISTER Solaiman Islam Age: 43 01936-631470  ঘরের বাইরে কোন মাক পরিদর্শন করুন কীট/অণুজীবমুক্ত এড়িয়ে চলুন, একে ঘরের বোকে ও খুঁচু খুঁচু বজায় রাখুন কিছুক্ষণ পর পর কমপক্ষে ২০ সেকেন্ড ঘরে সাবান পানি দিয়ে হাত ধুতে হবে হ্যাঁ না	← CC+ REGISTER Name: Gofur Ahmed ID: 01922333444 Today, 28 April brac Served 20 Members 50 CC visit Visit List	← PH REGISTER Solaiman Islam Age: 43 01936-631470  পিআইআই-এর শরীরের তাপমাত্রা হ্যাঁ না	← PH REGISTER Solaiman Islam Age: 43 01936-631470  গত ২৪ ঘণ্টায় পিআইআই-এর জ্বর ছিল কি? হ্যাঁ না
← HH REGISTER Solaiman Islam Age: 43 01936-631470  পরিবারের কতজন সদস্যের উচ্চরক্তচাপ/ ডায়াবেটিস রয়েছে? No. of people পূর্ববর্তী পরবর্তী	← HH REGISTER Solaiman Islam Age: 43 01936-631470  ঘরের বাইরে কোন মাক পরিদর্শন করুন কীট/অণুজীবমুক্ত এড়িয়ে চলুন, একে ঘরের বোকে ও খুঁচু খুঁচু বজায় রাখুন কিছুক্ষণ পর পর কমপক্ষে ২০ সেকেন্ড ঘরে সাবান পানি দিয়ে হাত ধুতে হবে হ্যাঁ না	← CMED HEALTH User ID: 01677080481-04 Name: _____ Date of Birth: _____ Age: _____ Gender: _____ REGISTER CANCEL	← PH REGISTER Solaiman Islam Age: 43 01936-631470  পিআইআই-এর শরীরের তাপমাত্রা হ্যাঁ না	← PH REGISTER Solaiman Islam Age: 43 01936-631470  গত ৬ ঘণ্টায় পিআইআই কি শ্বাসকষ্টজনিত/ এস/দশা জারীয়ে রাখা হয়েছে কি না? হ্যাঁ না

The flowchart illustrates the user journey for the CMED Health app, divided into five columns representing different stages: Registration, Profile Setup, Service Selection, Screening, and Results. Each stage shows a sequence of screens with icons, text, and buttons.

Column 1: Registration

- Screen 1: "PII REGISTER" header, icon of two people, text: "ঘরের বাইরে গেলে মাস পরিধান করুন", "জীভ/অন্যমাংসমাংস এড়িয়ে চলুন, একে আসলে থেকে ও খুঁচি দূরত্ব বজায় রাখুন", "কিছুক্ষণ পর পর কমপক্ষে ২০ সেকেন্ড ধরে সাবান পানি দিয়ে হাত ধুতে হবে", "জিও-হেলথ" button.
- Screen 2: "PII REGISTER" header, icon of two people, text: "ঘরের বাইরে গেলে মাস পরিধান করুন", "জীভ/অন্যমাংসমাংস এড়িয়ে চলুন, একে আসলে থেকে ও খুঁচি দূরত্ব বজায় রাখুন", "কিছুক্ষণ পর পর কমপক্ষে ২০ সেকেন্ড ধরে সাবান পানি দিয়ে হাত ধুতে হবে", "এসপি ডক" button.
- Screen 3: "PII REGISTER" header, icon of two people, text: "ঘরের বাইরে গেলে মাস পরিধান করুন", "জীভ/অন্যমাংসমাংস এড়িয়ে চলুন, একে আসলে থেকে ও খুঁচি দূরত্ব বজায় রাখুন", "কিছুক্ষণ পর পর কমপক্ষে ২০ সেকেন্ড ধরে সাবান পানি দিয়ে হাত ধুতে হবে", "জিও-হেলথ" button.
- Screen 4: "PII REGISTER" header, icon of two people, text: "ঘরের বাইরে গেলে মাস পরিধান করুন", "জীভ/অন্যমাংসমাংস এড়িয়ে চলুন, একে আসলে থেকে ও খুঁচি দূরত্ব বজায় রাখুন", "কিছুক্ষণ পর পর কমপক্ষে ২০ সেকেন্ড ধরে সাবান পানি দিয়ে হাত ধুতে হবে", "এসপি ডক" button.

Column 2: Profile Setup

- Screen 1: "PII REGISTER" header, icon of two people, text: "Solaiman Islam", "Age: 43", "01936-631470", "এসপি ডক" button.
- Screen 2: "PII REGISTER" header, icon of a person coughing, text: "পরিবারের অন্য কোন সদস্যের গত ৭ দিনে জ্বর, কাশি, শ্বাসকষ্টজনিত সমস্যা হয়েছে কিনা?", "হ্যাঁ" button, "না" button.
- Screen 3: "PII REGISTER" header, icon of a person coughing, text: "পরিবারের অন্য কোন সদস্যের গত ৭ দিনে জ্বর, কাশি, শ্বাসকষ্টজনিত সমস্যা হয়েছে কিনা?", "হ্যাঁ" button, "না" button.
- Screen 4: "PII REGISTER" header, icon of a person coughing, text: "পরিবারের অন্য কোন সদস্যের গত ৭ দিনে জ্বর, কাশি, শ্বাসকষ্টজনিত সমস্যা হয়েছে কিনা?", "হ্যাঁ" button, "না" button.

Column 3: Service Selection

- Screen 1: "SC REGISTER" header, icon of a person coughing, text: "আপনি কি কোনো পরীক্ষা করেছেন?", "হ্যাঁ" button, "না" button.
- Screen 2: "SC REGISTER" header, icon of a person coughing, text: "BRAC Mobile Team", "Govt Mobile Team", "Govt Facilities", "Private Health Facilities", "পূর্ববর্তী" button, "শুধু নতুন" button.
- Screen 3: "SC REGISTER" header, icon of a person coughing, text: "আপনি কি কোনো পরীক্ষা করতে ইচ্ছুক?", "হ্যাঁ" button, "না" button.
- Screen 4: "SC REGISTER" header, icon of a person coughing, text: "Positive", "Negative", "Waiting For Result", "পূর্ববর্তী" button, "শুধু নতুন" button.

Column 4: Screening

- Screen 1: "SC REGISTER" header, icon of a person coughing, text: "Solaiman Islam", "Age: 43", "01936-631470", "এসপি ডক" button.
- Screen 2: "SC REGISTER" header, icon of a person coughing, text: "কোনো পরীক্ষার ফলাফল কি?", "Positive", "Negative", "Waiting For Result", "পূর্ববর্তী" button, "শুধু নতুন" button.
- Screen 3: "SC REGISTER" header, icon of a person coughing, text: "আপনি কি কোনো পরীক্ষা করতে ইচ্ছুক?", "হ্যাঁ" button, "না" button.
- Screen 4: "SC REGISTER" header, icon of a person coughing, text: "আপনার পরামর্শ অনুযায়ী এসপি কে কি মাসপাতলা বায়োর পরামর্শ দেওয়া হয়েছে?", "হ্যাঁ" button, "না" button.

Column 5: Results

- Screen 1: "CMED HEALTH" header, "Serve Now" button, "CMED CallPhone Number: 0167-70804", list of users: "KORIF Ullah Age: 41", "Shohid Ali Age: 41", "Sharifa Begum Age: 24", "Md Towhid Age: 26", "Create a new user under this Mobile Number" button.
- Screen 2: "SCREENED LIST" header, icon of a person coughing, text: "Md Towhid", "Age: 26", "01911333666", "Visit Date: 18 Nov 2020", "স্ক্রিনেড তালিকা" button.
- Screen 3: "VISIT LIST" header, "Visit Date: 15 Sep 2020" button, list of users: "Md Towhid Age: 26 01911333666", "Md Towhid Age: 26 01911333666", "Md Towhid Age: 26 01911333666", "০০০০ ডক" button.